

This is one of PPM versions for running multiple PDB files. It works in Unix/Linux environment. This is not exactly the same version as implemented in PPM web server. The distribution includes:

1. Source code files (.f) and Makefile
2. Amino acid residue library, res.lib
3. Examples of input files (test.inp, 1rsy.pdb and 1gzm.pdb) and output files produced with such input files (RES1, datapar1, datasub1, 1rsyout.pdb and 1gzmout.pdb)

Compilation of executable. You will need **gfortran** compiler (use “make” command- see **Makefile**). The example of Makefile uses “fortran-9”. You should replace it to fortan version existing in your system (typically in directory ./usr/bin). Makefile will produce an executable named “immers”. There could be multiple warnings during compilation. Do not pay attention as far as the executable “immers” has been generated.

Running: immers<**test.inp**>name_of_output_file

Everything is assumed to run in the same directory. Make sure that file **res.lib** is located in the same directory.

Input. See an example of **test.inp** file. 0 or 1 in each line means “do not use” or “use” heteroatoms in the input PDB file (solvent molecules are always excluded), “in” or “out” means topology of N-terminus of first subunit included in the corresponding input PDB file.

Do not pay attention to “floating-point exceptions” message during execution if it happens (this is because PPM was developed with an earlier version of Fortran)

Output. The program produces output files named “datapar1” and “datasub1” for the set of input PDB files. They include calculated hydrophobic thickness/immersion depth, tilt angles, etc.

datapar1:

1gzm.pdb ;31.8; 1.6; 11; 0; -77.1;

means that hydrophobic thickness/depth is 31.8±1.6 Å, tilt angle is 11±0 degrees, and transfer energy is -77.1. kcal/mol

datasub1 (it includes only proteins identified as transmembrane by PPM):

1gzm.pdb ;A;11; 1(35- 60), 2(72- 99), 3(109- 133), 4(151- 173), 5(202- 224), 6(253- 277), 7(286- 308)

provides name of subunit, calculated tilt angle of the subunit and TM segments.